

Exam Date & Time: 05-May-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL / MAY 2024
SUBJECT: CHM 1071-CHM/CHM 1071-B/CHM 1051-B - ENGINEERING CHEMISTRY
ENGINEERING CHEMISTRY [CHM 1071-CHM]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) What is meant by ion-selective electrode? Derive an expression for electrode potential of glass electrode and write any two merits of it. (4)
- 1B) Distinguish between the following:
i) Activation and Concentration polarization
ii) Electroplating and Electroless plating (4)
iii) Fuel Cell and Galvanic Cell
iv) Primary and secondary batteries
- 1C) A galvanic cell consists of copper plate immersed in 10 M solution of CuSO_4 and iron plate immersed in 1 M FeSO_4 at 298 K. Write the cell reaction and calculate E.M.F. of the cell. (2)
- 2A) i) The concentration of yeast t-RNA in an aqueous solution is 10 M. The absorbance is found to be 0.209 when this Solution is placed in a 1.00 cm cuvette and 258 nm radiations are passed through it.
a) Calculate the specific absorptivity, including units, of yeast t-RNA. and
b) What will be the absorbance if the solution is 5 M? (4)
- ii) Explain the conductometric titration of weak acid Vs weak base with illustration. (2+2 = 4 marks)
- 2B) Define reverse osmosis. Describe with a neat diagram how sea water is purified using this technique. Write any two advantages of it. (4)
- 2C) i) Define electrodialysis.
ii) Distinguish between hard water and soft water (Any two points). (2)
- 3A) Discuss the autocatalytic nature of pitting corrosion with a suitable example. Explain any one primary and secondary factor each affecting the rate of pitting corrosion. (4)
- 3B) Differentiate (Any two points) between anodic and cathodic protection methods of corrosion control. Explain how use of anodic and cathodic inhibitor helps in decreasing the rate of corrosion. (4)
- 3C) Give reason for the following; (2)

- i) The position of a metal is fixed in electrochemical series but not in galvanic series.
- ii) Bolt and nut made of same metal is preferred in practice.

- 4A) Differentiate between the following (Any two points each) -
- i) Nematic and smectic phases
 - ii) Thermotropic and lyotropic liquid crystal. (4)
 - iii) Physical vapour deposition and chemical vapor deposition techniques.
 - iv) Top down method and bottom - up approach.

- 4B) i) Polyethylene sample has the following composition.

Degree of polymerization	400	300	200	100
% of composition	10	20	30	40

(4)
Calculate the number average molecular weight, weight average molecular weight and PDI of the given polymer sample. **Given:** Molecular weight of monomer is 28.

- ii) What is the effect of plasticizer on glass transition temperature of polymer?
(3+1 = 4 marks)

- 4C) i) Why polyvinyl chloride (PVC) has higher strength than Polyethylene?
ii) Write any two factors that affect the properties of composite materials. (2)

- 5A) Compute, compare and comment on the accuracy of method of determining the pH of an unknown solution in the following cases.
When a saturated calomel electrode ($E = 0.2422$ V) connected to a hydrogen electrode which is dipped in a solution of unknown pH, showed a potential of 0.39 V (4)
When the saturated calomel electrode connected to a glass electrode which is dipped in a buffer solution of pH = 2.2 gave an emf of 0.24 V and with the solution of unknown pH gave an emf of 0.26 V.

- 5B) i) List out the various steps involved in the sol-gel method of nano material synthesis.
ii) Distinguish (Any 4 points) between particle reinforced and fiber reinforced composites. (4)

- 5C) Describe the mechanism of wet corrosion (2)

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